Introductory Calculus

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The derivative as a limit**

Write the difference-quotient definition of $f'(x)$.

2 **[Foundational]** • **Target: The derivative as a limit**

Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.

3 **[Foundational]** • **Target: Differential equations**

Explain why solving a differential equation gives a family of functions rather than a single number.

4 **[Foundational]** • **Target: Differential equations**

Evaluate the indeterminate limit: $\lim_{x \to 5} \frac{x^2 - 25}{x^2 - 8x + 15}$

5 **[Foundational]** • **Target: Direct substitution**

Evaluate the limit at infinity involving a square root in the numerator: $\lim_{x \to \infty} \frac{\sqrt{9x^2 + 4x}}{2x - 1}$

6 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Evaluate the trigonometric limit: $\lim_{x \to 0} \frac{\sin(5x)}{\tan(2x)}$

7 **[Foundational]** • **Target: The derivative as a limit**

Evaluate the limit using algebraic conjugate rationalization: $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Differential equations**

Evaluate the limit of the rational expression by factoring: $\lim_{x \to 3} \frac{x^3 - 27}{x^2 - 2x - 3}$

9 **[Intermediate]** • **Target: Direct substitution**

Solve the nonlinear system of equations: $\{y = x^2 - 4x + 3; 2x + y = 6$ Find all points of intersection (x, y).

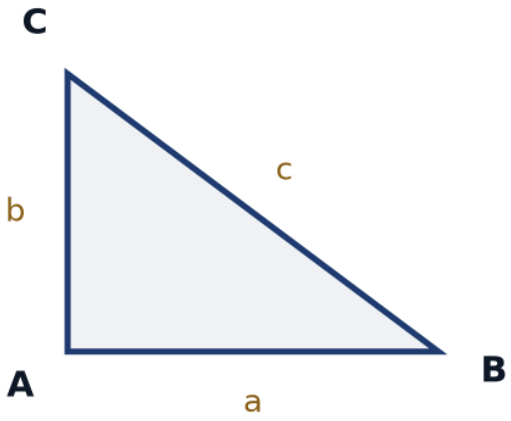
10 **[Intermediate]** • **Target: Indeterminate forms: factor and cancel**

Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{5x^3 - 2x + 7}{2x^3 + 4x^2 - 1}$

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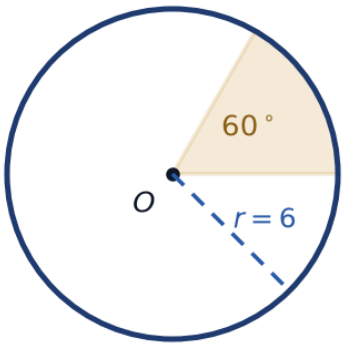
11 **[Intermediate]** • **Target: The derivative as a limit**

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



12 **[Intermediate]** • **Target: Differential equations**

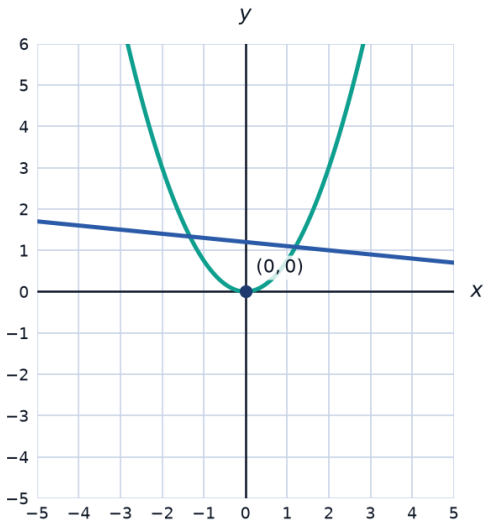
Find the modulus and principal argument of $z = -2\sqrt{3} + 2i$, and express z in polar form $r(\cos\theta + i\sin\theta)$.



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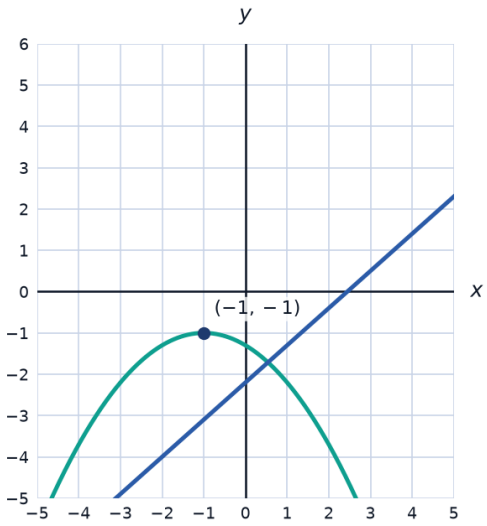
13 [Intermediate] • Target: Direct substitution

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



14 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find the distance in the polar coordinate plane between the two points $P_1\left(4, \frac{\pi}{3}\right)$ and $P_2\left(7, \frac{2\pi}{3}\right)$.

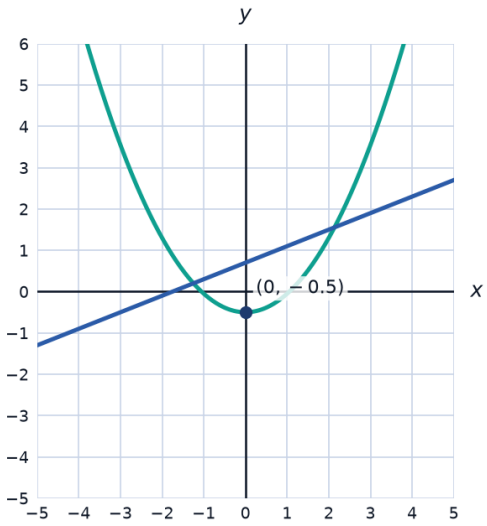


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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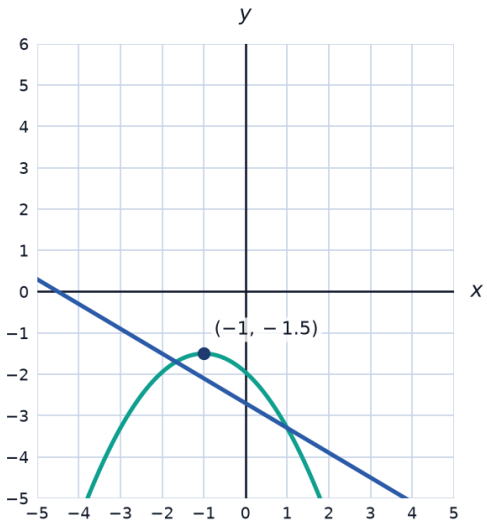
15 **[Challenging]** • **Target: The derivative as a limit** • **Cross-Lesson: 4.01 Matrices & Vectors**

Convert the polar coordinates $\left(6, \frac{5\pi}{4}\right)$ into rectangular Cartesian coordinates (x, y).



16 **[Challenging]** • **Target: Differential equations** • **Cross-Lesson: 6.01 Probability**

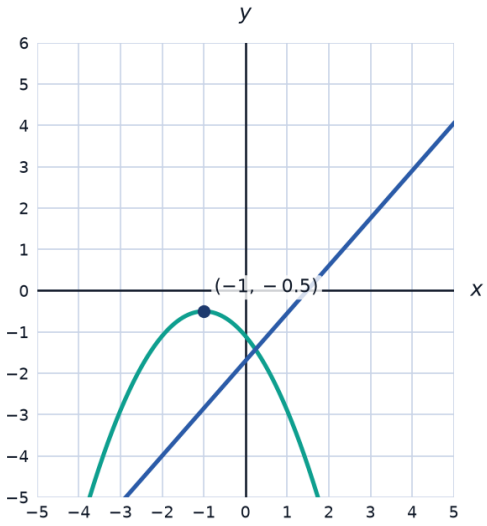
Convert the Cartesian equation $x^2 + (y - 4)^2 = 16$ into a polar equation relating r and θ.



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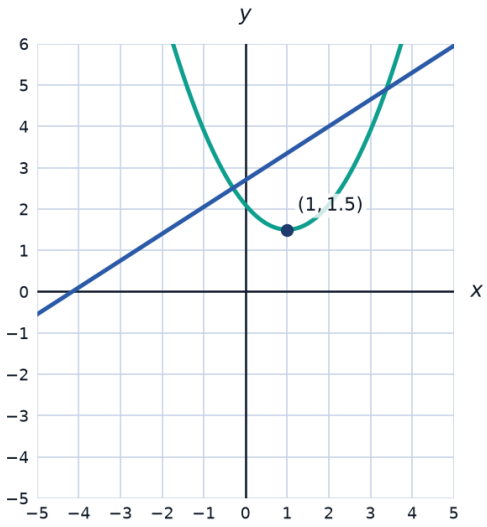
17 [Challenging] • Target: Direct substitution • Cross-Lesson: 1.01 Complex Numbers

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



18 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Sequences & Series

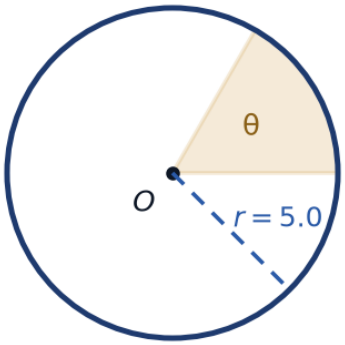
Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



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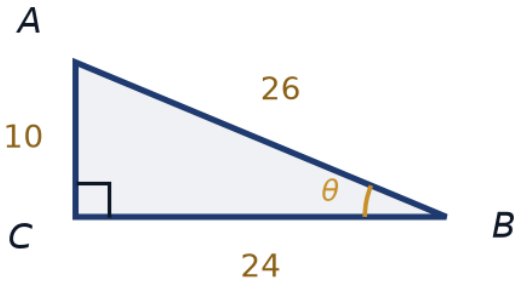
19 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 3.01 Functions & Graphs

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



20 [Challenging] • Target: Differential equations • Cross-Lesson: 4.01 Matrices & Vectors

Find the exact value of $\sin 30^\circ + \cos 60^\circ$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Direct substitution

Given $f(x) = -x^2 + 5x - 4$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(2)$.

a. 1

b. -1

c. 3

d. 4

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22 [Foundational] • Target: Indeterminate forms: factor and cancel

Find the average rate of change of $f(x) = x^2$ on the interval from $x = -1$ to $x = 0$.

a. -3

b. 3

c. 1

d. -1

23 [Foundational] • Target: The derivative as a limit

If $f(x) = 7x - 4$, find $f'(x)$ using the difference-quotient definition.

a. 7

b. -4

c. 8

d. 0

24 [Foundational] • Target: Differential equations

How many ways can 6 items be arranged, taken 2 at a time (order matters)?

a. 30

b. 15

c. 32

d. 720

25 [Foundational] • Target: Direct substitution

A fair coin is flipped 3 times. Find $P(\text{all heads})$.

a. $\frac{1}{8}$

b. $\frac{1}{6}$

c. $\frac{3}{8}$

d. $\frac{1}{4}$

26 [Foundational] • Target: Indeterminate forms: factor and cancel

Independent events A, B have $P(A) = \frac{4}{9}$ and $P(B) = \frac{1}{3}$. Find $P(A \text{ and } B)$.

a. $\frac{4}{27}$

b. $\frac{7}{9}$

c. $\frac{6}{41}$

d. $\frac{7}{18}$

27 [Foundational] • Target: The derivative as a limit

A fair six-sided die is rolled. Find $P(\text{rolling a } 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{5}{6}$

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28 [Foundational] • Target: Differential equations

A committee of 3 men and 1 women is to be chosen from 6 men and 5 women. How many ways?

a. 100

b. 25

c. 330

d. 106

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Direct substitution

A card is drawn from a standard 52-card deck. Find $P(\text{face card (J, Q, or K)})$.

a. $\frac{3}{13}$

b. $\frac{3}{52}$

c. $\frac{1}{13}$

d. $\frac{1}{4}$

30 [Intermediate] • Target: Indeterminate forms: factor and cancel

A bag has 5 black marbles and 7 white marbles. One is drawn at random. Find $P(\text{black})$.

a. $\frac{5}{12}$

b. $\frac{7}{12}$

c. $\frac{5}{7}$

d. $\frac{1}{2}$

31 [Intermediate] • Target: The derivative as a limit

A bag has 5 green marbles and 2 yellow marbles. One is drawn at random. Find $P(\text{green})$.

a. $\frac{5}{7}$

b. $\frac{2}{7}$

c. $\frac{5}{2}$

d. $\frac{1}{2}$

32 [Intermediate] • Target: Differential equations

A bag has 2 red marbles and 7 blue marbles. One is drawn at random. Find $P(\text{red})$.

a. $\frac{2}{9}$

b. $\frac{7}{9}$

c. $\frac{2}{7}$

d. $\frac{1}{2}$

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33 [Intermediate] • Target: Direct substitution

How many ways can a group of 2 be chosen from 4 people (order doesn't matter)?

a. 6

b. 12

c. 8

d. 4

34 [Intermediate] • Target: Indeterminate forms: factor and cancel

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?

a. 4

b. 3

c. 1

d. 6

35 [Intermediate] • Target: The derivative as a limit

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

a. 2

b. 8

c. 18

d. -30

36 [Intermediate] • Target: Differential equations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 - x + 4$.

a. $4x^2 - 1$

b. $8x + 4$

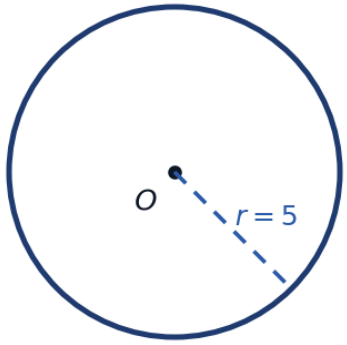
c. $4x - 1$

d. $8x - 1$

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37 [Intermediate] • Target: Direct substitution

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. 1

d. $\frac{\sqrt{2}}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Sequences & Series

What is the minimum value of the function $f(x) = \sqrt{x^2 - 4x + 13} + \sqrt{x^2 - 14x + 130}$ for all real numbers x?

a. 12

b. $4\sqrt{10}$

c. 13

d. $\sqrt{170}$

39 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 3.01 Functions & Graphs

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

a. -2

b. 2

c. 7

d. 16

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40 [Challenging] • Target: Differential equations • Cross-Lesson: 4.01 Matrices & Vectors

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

41 [Challenging] • Target: Direct substitution • Cross-Lesson: 6.01 Probability

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

a. $(1, 1)$

b. $(-5, 1)$

c. $(1, 9)$

d. $(-5, 9)$

42 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 1.01 Complex Numbers

If the limit $\lim_{x \to 3} \frac{x^2 + kx - 15}{x^2 - 4x + 3} = L$ exists and is a finite real number, where k and L are constants, what is the value of $k + L$?

a. 2

b. 4

c. 6

d. 8

43 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 2.01 Sequences & Series

For each real number $t \in (-2, 2)$, the cubic equation $x^3 - 3x + t = 0$ has three distinct real roots, denoted in increasing order by $x_1(t) < x_2(t) < x_3(t)$. What is the value of the limit $\lim_{t \to 2^-} \frac{x_3(t) - x_2(t)}{\sqrt{2 - t}}$?

a. $\frac{2\sqrt{3}}{3}$

b. $\sqrt{3}$

c. $\frac{\sqrt{3}}{2}$

d. $\frac{4}{3}$

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44 [Challenging] • Target: Differential equations • Cross-Lesson: 3.01 Functions & Graphs

What is the value of the limit $\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2}$?

a. 5

b. 1

c. -1

d. Does not exist

45 [Challenging] • Target: Direct substitution • Cross-Lesson: 4.01 Matrices & Vectors

The function f is defined by $f(x) = \frac{3x + 2}{x - 4}$ for all real numbers $x \neq 4$. What is the value of $f^{-1}(2)$?

a. -10

b. -6

c. -4

d. 6

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